

Gokul Krishna C K

 github.com/piston-head  [+91-7907435231](tel:+91-7907435231)  linkedin.com/in/YOUR-LINKEDIN  gokulkrishna1282@gmail.com

SUMMARY

Backend Engineer with experience building scalable microservices using Spring Boot and Java. Skilled in designing event-driven systems and developing healthcare-compliant APIs using FHIR R4. Focused on building reliable, secure, and production-ready backend systems.

EXPERIENCE

Philips

2023 – Present

Software Engineer I → Software Engineer II

- **Data API Service:** Designed and developed a Spring Boot GraphQL service to transform healthcare data into FHIR R4-compliant resources
- Integrated with Rhapsody engine to generate and validate GraphQL queries for healthcare data exchange
- Implemented data transformation using HAPI FHIR with validation to ensure interoperability compliance
- Implemented Traefik as a reverse proxy and load balancer to distribute incoming traffic across services
- **Github Issue Ops automation:** Designed and implemented GitHub Actions-based automation to process structured issue requests and trigger cross-repository workflows
- Built onboarding and offboarding workflows using label-based locking to prevent race conditions in concurrent executions
- Automated infrastructure lifecycle tasks including ArgoCD cleanup, Kubernetes namespace deletion, and GitOps synchronization
- Developed comment-driven command parsing and approver validation for workflow execution through GitHub issues

PROJECT

Conversational RAG Application

2025-2026

- Built an end-to-end RAG pipeline parsing multi-format documents (PDF, DOCX, TXT) via Apache Tika, generating local 384-dim embeddings with ONNX embedding model, and retrieving semantically relevant chunks via cosine similarity to produce grounded, hallucination-resistant answers using Google Gemini 2.5 Flash.
- Implemented server-side conversational memory with a thread-safe session store and sliding 5-turn history injected as structured User message turning it into the Gemini prompt, transforming a stateless Q&A system into a context-aware multi-turn conversational agent.
- Secured the API with stateless JWT authentication with role-based access control, and a global exception handler surfacing Gemini API errors as structured HTTP responses.

SKILLS AND EXPERTISE

Backend & APIs: Java, Spring Boot, Spring Data JPA, Microservices, Spring Security (JWT, OAuth2), REST, GraphQL

Messaging & Healthcare: Apache Kafka, Resilience4j, FHIR R4, Rhapsody Integration Engine

Databases & DevOps: MS SQL Server, Firebase, Docker, GitHub Actions, Fortify, JUnit, Mockito

ACHIEVEMENTS AND CERTIFICATIONS

- AWS Certified Cloud Practitioner (CLF-C02)
- Google Certification: Introduction to Generative AI

EDUCATION

Amrita Institute of Technology

Bachelor of Technology in Computer and Communication Engineering

2019 – 2023

St. Jude's Public School and Junior College

Council for the Indian School Certificate Examinations

2015 – 2019